

Consulting Final Project



The Ohio State University

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Overall Goal

- Fisher's iris data with measurements in centimeters of sepal length and width and petal length and width for three species of iris
- Identify whether there is statistically significant difference in features for different species
- Build a model that can predict the species based on their features

Testing difference in features

- By the EDA conducted by the client, the distribution of features differ across different species
- Conduct ANOVA to identify statistical significance of this difference
- Let Y_{it} be the value of each feature (e.g. sepal length) for t^{th} observation of species i

$$Y_{it} = \mu + \tau_i + \epsilon_{it}$$

- ANOVA tests whether τ_i are the same, i.e., $H_0 : \tau_1 = \tau_2 = \tau_3$

	Sepal Length	Petal Length	Petal Width
F-statistic	119.3	1180	960
p-value	≈ 0	≈ 0	≈ 0

Testing difference in features

- We can quantify the difference of each feature between species. For example, for sepal length,

Sepal Length	Estimate	Std. Error	p-value
Versicolor	0.93	0.103	8.7×10^{-16}
Virginica	1.58	0.103	2.2×10^{-16}

- We can conclude that compared to setosa, the average sepal lengths of versicolor and virginica are 0.93cm and 1.58cm longer, consistent with the result provided by the EDA

Testing difference in features

- Results for petal lengths and widths are provided below
- Compared to setosa, the petal lengths of versicolor and virginica are 2.79cm and 4.09cm longer
- The petal width of versicolor and virginica are 1.08cm and 1.78cm wider

Petal Length	Estimate	Std. Error	p-value
Versicolor	2.79	0.086	≈ 0
Virginica	4.09	0.086	≈ 0

Petal Width	Estimate	Std. Error	p-value
Versicolor	1.08	0.04	≈ 0
Virginica	1.78	0.04	≈ 0

Principal Component Analysis

- As noted from the EDA, the first two components capture 95% of the total variability, but have significant overlap between versicolor and virginica.
- We test whether we can come up with a summary statistic of the original data that statistically differ between species using PCA

Principal Component Analysis

- PCA gives us a linear combination of the original variables that capture the most variances among them
- The first principal component is approximately proportional to Sepal Length + Petal Length + Petal Width - 1/2 Sepal Width

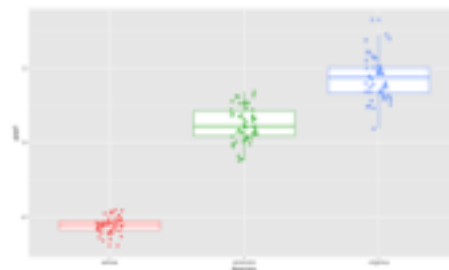


Figure 1: First Principal Component

Principal Component Analysis

- Again conduct ANOVA but with the first principal component as the response variable
- We arrive at the conclusion that there is a significant difference in the first principal component between the species

Sepal Length	
F-statistic	1051
p-value	≈ 0

Classification

- Conduct logistic regression with features(X) as predictors and the species as the response(Y)

$$\log \frac{P(Y = \text{setosa})}{1 - P(Y = \text{setosa})} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- `vglm(Species ~ Sepal.Length + Sepal.Width + Petal.Length + Petal.Width, family = multinomial, data = iris)`

	setosa	versicolor	virginica
setosa	50	0	0
versicolor	0	49	1
virginica	0	1	49

Classification

- A more robust classification can be done by dividing the dataset into training and test set
- Fit the logistic model using the training set and conduct prediction using the test set
- Use 100 observations for training and 50 observations for test

	setosa	versicolor	virginica
setosa	17	0	0
versicolor	0	17	0
virginica	0	1	15